

**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025

OPERATING SYSTEMS

(COMMON TO IT,CSE(AI&ML),CSE(DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) What is Operating Systems? List any two of its main goals. [1M] L1
- b) Write any two differences between tar and gzip commands in Linux. [1M] L1
- c) What is a Process Control Block (PCB) ? List its key components. [1M] L1
- d) What are the types of threads? Mention any two. [1M] L1
- e) Explain IPC using unnamed and named pipes. [1M] L2
- f) Write Short notes on the Resource Allocation Graph (RAG). [1M] L2
- g) What is Virtual Memory? How does it improve memory utilization. [1M] L2
- h) Explain the concept of Domain Structure in protection mechanism. [1M] L2
- i) Define a File and mention its various operations. [1M] L1
- j) How does link() differ from symlink()? [1M] L2

PART- B

(5 X 10 =50 Marks)

2. a) Explain the various operations provided by an Operating System. [5M] L2
- b) List and explain the major features of the Linux Operating System. [5M] L2

OR

3. a) With diagrams, explain Simple and Layered Operating System structures along with their pros and cons. [5M] L4
- b) Draw the Linux architecture and explain each layer briefly with a neat diagram. [5M] L3
4. a) Explain the concept of context switching in a multitasking operating system. Support your answer with a neat diagram. [5M] L2
- b) Solve the Dining Philosophers Problem using Semaphores. Justify how your solution ensures synchronization without deadlock. [5M] L4

OR

5. a) What are process schedulers? Describe long-term, short term and medium term schedulers with examples. [5M] L2

b) Explain the types of CPU scheduling algorithms and describe the scheduling criteria used to evaluate them. [5M] L3

6. a) Describe how message queues are used in IPC. Explain all system calls like `msgget()`, `msgsnd()`, `msgrcv()` and `msgctl()` with syntax. [5M] L2

b) Write the steps of the Safety Algorithm used in the Banker's Algorithm. Explain how it determines a system's safety. [5M] L3

OR

7. Apply the snapshot below, determine whether process P1's request for 1 instance of A and 2 instances of C can be safely granted. Apply the Resource Request Algorithm. Show all steps. [10M] L3

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

8. a) Define fragmentation in memory management. Differentiate between internal and external fragmentation [5M] L2

b) Describe the goals and principles of protection in operating systems. Why is memory and process protection necessary? Give examples [5M] L2

OR

9. Consider the following reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, apply FIFO, Optimal and LRU Page replacement algorithms. Memory has 4 frames. Calculate Page Faults for each method and analyze which is best. [10M] L3

10. a) Explain the various file allocation methods. Provide for each a neat diagram, advantages and disadvantages. [5M] L2

b) For the file related system calls `creat()`, `read()`, `open()`, `chmod()`, `lseek()`, explain their syntax parameter, return values, and examples in C. [5M] L3

OR

11. a) Discuss free space management techniques. Highlight how each works and compare their efficiency, pros and cons. [5M] L3

b) For the directory related system calls `opendir()`, `readdir()`, `umask()`, `mkdir()`, `rmdir()`, describe each with usage syntax, parameter list and example programs. [5M] L3



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OPERATING SYSTEMS

(COMPUTER SCIENCE AND ENGINEERING)

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Max. Marks: 60

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PART- A

(10 Marks)

1. a) List the types of computer software and Explain each in one line. [1M] L1
- b) Mention the purpose and syntax of cat and script commands in Linux. [1M] L2
- c) Compare the usage of fork() and vfork() system calls. [1M] L2
- d) What is a semaphore? Mention its types and purpose. [1M] L1
- e) What is Deadlock Recovery? Briefly explain any two methods of recovering from deadlock. [1M] L2
- ~~f) Define Safe State and Safe Sequence in the context of deadlock avoidance. [1M] L1~~
- g) Define Demand Paging and Page Fault with simple example. [1M] L2
- h) Define Swapping. How does it help in memory management? [1M] L1
- i) What is free space management in file systems and why is it needed? [1M] L1
- j) How does stat () differ from lstat()? [1M] L2

PART- B

(5 X 10 =50 Marks)

2. a) Describe the operational aspects of an operating system, focusing on Dual-mode operation and the use of Timers. [5 M] L2
- b) Explain the various services provided by an Operating System with simple example. [5M] L2

OR

3. Analyze the different operating system structures (Simple, Layered, Microkernel, Modular). Discuss their advantages and disadvantages in detail. [10 M] L4
4. Explain hardware-based synchronization methods using atomic instructions like Test-and-Set or Compare-and-swap. Provide pseudo-code examples. [10 M] L3

OR

5. For the following process, use Round Robin Scheduling with a time quantum of 2 times. Draw the Gantt chart and Calculate average waiting and turnaround time. [10M] L3

PID	Arrival Time (AT)	Burst Time (BT)
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

6. Consider the following snapshot of a system, Use the Banker's Algorithm to determine whether the system is in a safe state. If yes, provide the safe sequence. Show all steps. [10M] L3

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	1			
P4	0	0	2	4	3	3			

OR

7. a) Describe how shared memory is used in IPC. Explain all system calls like shmget(), shmat(), shmdt(), shmctl() with syntax. [5M] L3
- b) Explain how a Wait-for Graph is used to detect Deadlocks when each resource type has only a single Instance. [5 M] L2
- 8 a) Explain the two types of Contiguous Memory Allocation. Describe with Advantages and disadvantages. [5M] L2
- b) Explain the Access Matrix model used for protection. Describe how it is implemented. [5M] L3

OR

9. a) Describe the types of Linking and Loading with suitable examples. Explain their advantages and typical usage. [5 M] L3
- b) Explain address binding and describe the three types of bindings. Show how address binding changes across these stages. [5M] L2
- 10 a) Discuss the importance of file protection in operating systems. Describe access control methods. [5 M] L3
- b) Illustrate the Layered Architecture of the file system. [5 M] L2
- OR
11. a) Explain the three file access methods. For each provide a real-time use case diagram, and compare advantages and disadvantages. [5M] L2
- b) Describe different directory structures. Include advantages, disadvantages illustrative diagrams. [5M] L2

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Time: 3 hours

Max. Marks: 70

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Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

1. a) What are the main functions of an operating system? (20 Marks)
- b) Distinguish between kernel mode and user mode. [2M] L1
- c) What are the different states of a process? [2M] L2
- d) What is a context switch? [2M] L1
- e) Define semaphore. [2M] L2
- f) What are necessary conditions for deadlock? [2M] L1
- g) What is the difference between logical and physical address? [2M] L2
- h) What is a directory structure? [2M] L2
- i) What is access matrix? [2M] L1
- j) Define domain of protection. [2M] L2

PART- B

2. Describe the architecture and key features of LINUX. Elaborate on the functionality of at least 5 essential LINUX commands with examples. (5 X 10 =50 Marks)
- [10M] L2
- OR**
3. Illustrate how system calls work with an example. [10M] L3
4. Describe the process states and Process Control Block (PCB). [10M] L2
- OR**
5. Explain any three CPU scheduling algorithms with examples. [10M] L2
6. Explain how Semaphores are used for IPC, including the associated system calls. [10M] L2
- OR**
7. Explain interprocess communication using message queues and shared memory. [10M] L2

8. Compare various page replacement algorithms with an example each.

[10M] L2

OR

9. Explain different File Allocation Methods, including Contiguous Allocation, Linked Allocation, and Indexed Allocation, along with Free-space Management techniques.

[10M] L2

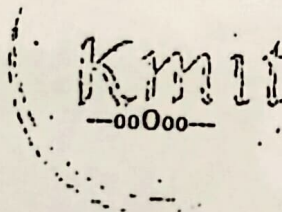
10. Explain the Goals and Principles of Protection in operating systems.

[10M] L2

OR

11. Illustrate how protection mechanisms are implemented using access matrix.

[10M] L2





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OPERATING SYSTEMS

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours

Max. Marks: 70

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Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) Define Operating System. [2M]L1
- b) Distinguish the multi programming and multi-processing. [2M]L2
- c) What are the two operations on process? [2M]L1
- d) Define the Deadlocks. [2M]L1
- e) What is a semaphore? Why it is used? [2M]L1
- f) Discuss about the message queues. [2M]L2
- g) Write a short note on Segmentation. [2M]L1
- h) Write about demand paging. [2M]L1
- i) Explain about socket and socket addresses. [2M]L2
- j) List out the principles of Protection. [2M]L1

PART- B

(5 X 10 =50 Marks)

2. a) Explain about the structure of OS in detail. [5M]L2
 - b) What are the different types of operating systems? Explain them in detail. [5M]L2
- OR
3. List out various system components in OS. Explain about them. [10M]L2
 4. a) What are the advantages and disadvantages of using the system call interface? [5M]L2
 - b) Following is the snapshot of a [5M]L3

CPU Process	CPU Burst	Arrival Time
P1	5	0
P2	9	2
P3	3	3
P4	7	4

Show the Gantt chart and calculate the turnaround time and waiting time of the jobs for FCFS.

OR

5. a) Write a short note on Deadlock Prevention and Deadlock avoidance. [5M]L2
 b) Choose 3 processes P1, P2 and P3, which require 7, 9 and 6 time units and arrive at times 0, 2 and 3. Draw the Gantt chart, calculate the process completion time and average waiting time for SJF. [5M]L3
6. a) Explain dining-philosopher problem with example. [5M]L2
 b) Discuss the facts about Critical Regions and Monitors. [5M]L2

OR

7. a) Explain about Critical Section Problem in detail. [5M]L2
 b) Demonstrate the IPC mechanism using FIFOs. [5M]L2

8. a) Explain in detail Contiguous Memory Allocation. [5M]L2
 b) Take the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Identify the no. of page faults that would occur for the optimal replacement algorithm, assuming three or four frames. Remember all frames are initially empty, so your first unique pages will all cost one fault each. [5M]L3

OR

9. a) Explain the terms in Memory Partitioning with examples: [5M]L2
 i) Fixed Partitioning ii) Dynamic partitioning
 b) Explain in detail about File system structure. [5M]L2
10. a) Explain the Domain of protection requirements. [5M]L2
 b) Discuss the final goals of Protection. [5M]L2

OR

11. a) Identify the major Socket attributes and Connection requirements. [5M]L2
 b) Discuss in detail about following terms [5M]L2
 i) Bind ii) listen iii) accept

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B.TECH II YEAR II SEMESTER REGULAR EXAMINATIONS, AUGUST - 2023

OPERATING SYSTEMS

(COMMON TO IT, CSE (DATASCIENCE) & CSE (AI&ML))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 = 20 Marks)

1. a) List out the operations of an operating system [2M]
- b) List the types of operating System structures [2M]
- c) What is a thread? [2M]
- d) List out methods for handling deadlock? [2M]
- e) Define binary semaphore? [2M]
- f) Write any two classical problems of synchronization [2M]
- g) Write system calls of create, open, write, close [2M]
- h) Write the difference between Physical and logical address space [2M]
- i) Write the goals of protection [2M]
- j) What are the socket attributes? [2M]

PART- B

(5 X 10 = 50 Marks)

2. a) Define Operating System. Explain simple layered approach of operating systems in detail. [5M]
- b) Describe about operating System services. [5M]

OR

3. a) Explain the architecture of LINUX operating system with a neat diagram and discuss its features. [5M]
 - b) Illustrate about Various File processing utilities, Text Processing utilities. [5M]
4. a) Define a Process. Describe Process state transition diagram with a near sketch. [5M]
 - b) Construct the Gantt chart for i) Shortest job first ii) Round Robin with $q=3$ iii) Round robin with $q=4$ iv) FCFS scheduling algorithms for the following.

Process	P1	P2	P3	P4	P5
Arrival time	0	0	2	1	3
CPU Burst Time (in ms)	10	6	12	8	5

[5M]

OR

P.T.O

5a) What are various system calls for process management?

[4M]

b)

Consider the following snapshot of a system:

Processes	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

a) What is the content of the matrix Need?

b) Is the system in a safe state?

[6M]

6. a) What is critical section problem? Explain about semaphores.

[5M]

b) Explain about producer consumer Problem in detail.

[5M]

OR

7. a) Define FIFOs. Write the syntax for Creation, Sending and Receiving information by using FIFOs. Discuss with clear examples.

[5M]

b) Differentiate how Inter process Communication happens through Semaphores and Message Passing.

[5M]

8. a) A process refers to 5 pages, A, B, C, D, and E in the order- A; B; C; D; A; B; E; A; B; C; D; E. If the page replacement algorithm is LRU, calculate the number of page faults with empty frames of size 4?

[5M]

b) Write short notes on: i) Demand Paging. ii) Segmentation.

[5M]

OR

9. a) Explain in detail about file Structures.

[5M]

b) Write short notes on a) opendir b) closedir c) mkdir d) rmdir e) readdir system calls

[5M]

10. a) What are the Principles of protection.

[5M]

b) Explain about domain of protection.

[5M]

OR

11. a) Explain about socket communications ?

[5M]

b) Describe about socket options & socket Connections?

[5M]

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PART- A

(2 X 10 =20 Marks)

1. a) What is an Operating system? [2M]
- b) List the Services of operating systems. [2M]
- c) What is a thread? [2M]
- d) List out various scheduling criteria for CPU scheduling. [2M]
- e) Define semaphore. [2M]
- f) What are the requirements that a solution to the critical section problem must satisfy? [2M]
- g) What is logical address space and physical address space? [2M]
- h) Define Demand paging. [2M]
- i) How security and protection services are managed by OS? [2M]
- j) What are the socket elements in OS? [2M]

PART- B

(5 X 10 =50 Marks)

2. a) What are various Operations of Operating System? [5M]
- b) Describe different types of Operating System structures. [5M]

OR

3. a) With a neat Diagram, Discuss the Architecture of LINUX Operating System? Recall its features? [5M]
- b) Illustrate about Various Linux utilities [5M]

4. a) Define Thread. What are the advantages and uses of threads? [5M]
- b) Distinguish between a thread and a process? [5M]
- b) Explain about Deadlock avoidance, Deadlock prevention, Deadlock Detection [5M]

OR

P.T.O

5. a) Consider the following set of process assumed to have arrived at time 0, in the order P1,P2,P3,P4,P5 with the length of the CPU burst given in milliseconds.
(Assume low numbers for high priority)

Process	Burst time	Priority
P1	10	3
P2	1	1
P3	2	4
P4	1	5
P5	2	5

- i) Using priority scheduling algorithm schedule the processes using a Gantt chart
ii) Find the average waiting time (Assume low numbers for high priority). [4M]
- b) Define process. How many different states a process has? Explain when a process changes the state with a state diagram? [6M]
6. a) What is critical section problem? Write and explain Peterson's solution for it. [5M]
b) Explain briefly about Classical synchronization problems. [5M]
- OR
7. a) Define Message Queues. Write the syntax for creation, sending and receiving information using message queues. [5M]
b) Explain about Semaphores, Pipes in IPC. [5M]
- 8.a) Consider the following page reference string 2,3,4,5,3,2,6,7,3,2,3,4,1,7,1,4,3,2,3,4,7. Calculate the number of page faults with LRU, FIFO and optimal page replacement algorithms with frame size of 3. [5M]
b) Discuss the procedure for page fault in demand paging. [5M]
- OR
9. a) Explain in detail about file system structure and implementation. [5M]
b) Write short notes on
a) read () b) write () c) lseek() d) stat e) chmod [5M]
10. a) Explain about Goals & Principles of protection. [5M]
b) Explain about domain protection mechanism in brief. [5M]
- OR
11. a) List out the 3 types of sockets? What is the use of socket in OS? [5M]
b) What are socket Connections? [5M]

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OPERATING SYSTEMS

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Part A is compulsory which carries 25 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1. a) Define an Operating System. Mention any two primary objectives. [2M] L1
- b) Differentiate between Monolithic and Microkernel architectures. [3M] L2
- c) Define a process. How is it different from a program? [2M] L1
- d) Using an example, explain how fork() creates a child process in Linux. [3M] L3
- e) Differentiate between pipes and FIFOs with examples. [2M] L2
- f) What is the purpose of a Resource Allocation Graph? [3M] L2
- g) Define logical and physical address space. [2M] L1
- h) What is fragmentation? Differentiate between internal and external fragmentation. [3M] L2
- i) Define directory structure. List any two types of directory structures. [2M] L1
- j) Discuss the role of access control in system protection? [3M] L2

PART- B

(5 X 10 =50 Marks)

2. a) Describe the architecture of a computer system. How does the OS interact with the hardware? [5M] L3
- b) Compare and contrast between Real-Time Systems and Distributed Systems. [5M] L3

OR

3. a) Explain the usage of mv, ls, cat, mkdir, and rmdir commands with suitable examples. [5M] L3
- b) Illustrate the differences in OS requirements between personal computer, mobile and parallel computing environments. [5M] L3
4. a) Explain the various scheduling criteria and their significance in CPU scheduling. [5M] L2
- b) Compare different system calls used for process management [5M] L4

OR

5. Given the following processes, schedule them using the Shortest Job First algorithm. Draw Gantt chart, determine the Completion, Turnaround, and Waiting Time for each process, and calculate the Average Waiting and Turnaround Time.

[10M] L3

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	1
P4	5	4

6. a) Describe in detail the mechanisms of Inter-Process Communication. Explain how message queues and shared memory differ in implementation. [5M] L2
- b) Explain how monitors are used in process synchronization. [5M] L3

OR

7. Consider the following snapshot of a system:

[10M] L4

	Allocation	Max	Available
	A B C	A B C	A B C
P0	1 1 2	4 4 3	2 1 0
P1	2 1 2	3 2 2	
P2	4 0 1	9 0 2	
P3	0 2 0	7 5 3	
P4	1 1 2	1 1 2	

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- a) What is the content of the matrix Need?
- b) Is the system in a Safe state?
- c) If a request from process P1 arrive for (1, 1, 0) can the request be granted immediately?

8. a) Explain Segmentation in detail with neat diagram.

[5M] L2

- b) Describe the structure of a page table and its role in address translation.

[5M] L3

OR

9. Consider the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, and 3. How many page faults would occur for the following page replacement algorithms, assuming four frames?

- a. LRU page replacement. b. FIFO page replacement. c. Optimal page replacement. [10M] L4

10. a) Explain various file protection mechanisms.

[5M] L3

- b) Describe the goals and principles of protection in modern operating systems.

[5M] L3

OR

11. Illustrate the use of system calls such as open(), read(), write(), stat(), and close() with a sample C program that performs basic file operations.

[10M] L3

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